2. A 14-year-old female comes to the clinic experiencing abrupt, involuntary motions of her arms and limbs within the last week. When you ask her to grip your hand, you note that her hand strength weakens intermittently. Her mother mentions that she had an episode of sore throat about a month ago that was left untreated. The presence of which of the following additional clinical findings would be sufficient to confirm the most likely diagnosis?

A. ○ Fever of 39.2°C (102.5°F)

B. ○ Joint pain without inflammation

C. ○ Prolonged PR interval on ECG

D. ○ Punctate skin hemorrhages

E. ● Ring-shaped rashes on the trunk or limbs

Hide Explanation

The correct answer is ring-shaped rashes on the trunk or limbs (E). Erythema marginatum and the Sydenham chorea exhibited by this patient are both major manifestations of acute rheumatic fever (ARF), as defined by the Jones criteria. Other major findings of the Jones criteria are arthritis, heart disease, and subcutaneous nodules. Fever of 39.2°C (102.5°F) (A), joint pain without inflammation (B), and prolonged PR interval on ECG (C) are minor manifestations and would require two or more to be present simultaneously in addition to one major criterion for the diagnosis. Punctate skin hemorrhages (D) are not found in ARF.

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- Ziad I. A noob's guide to ECG. October 29, 2011. https://www.slideshare.net/irfanziad1/ecg-9939328

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Define acute rheumatic fever (ARF) and rheumatic heart disease (RHD).~~

02 ☑ ~~Differentiate the clinical presentation of ARF and RHD.~~

03 ☑ ~~Explain the pathophysiology of ARF and RHD.~~

04 ☑ ~~Interpret clinical findings to properly diagnose ARF and RHD.~~

05 ☑ ~~Outline the management plan of ARF and RHD.~~

Objective Confidence: 5/5

What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Mitral Regurgitation and Mitral Valve Prolapse

Last updated February 20, 2026 17 min read ScholarRx

Adapted by: Diptiman Bose, Patrick Brooks, Phil Riley

Editors: Chelsey Deel, Jordan Finn, Russell Johnson

Adapted from: Mitral Regurgitation

Listen to this Brick

0:00 / 24:50 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the clinical presentation of Mitral Regurgitation and Mitral Valve Prolapse.
2. Explain the etiologies and pathophysiology of acute and chronic Mitral Regurgitation and Mitral Valve Prolapse.
3. Explain how itral Valve

Mitral regurgitation (MR) is the failure of the mitral valve to remain closed during systole, leading to regurgitation of blood from the left ventricle (LV) back into the left atrium (LA) with each contraction of the heart (Figure 2). This results in a systolic murmur, best heard at the apex.

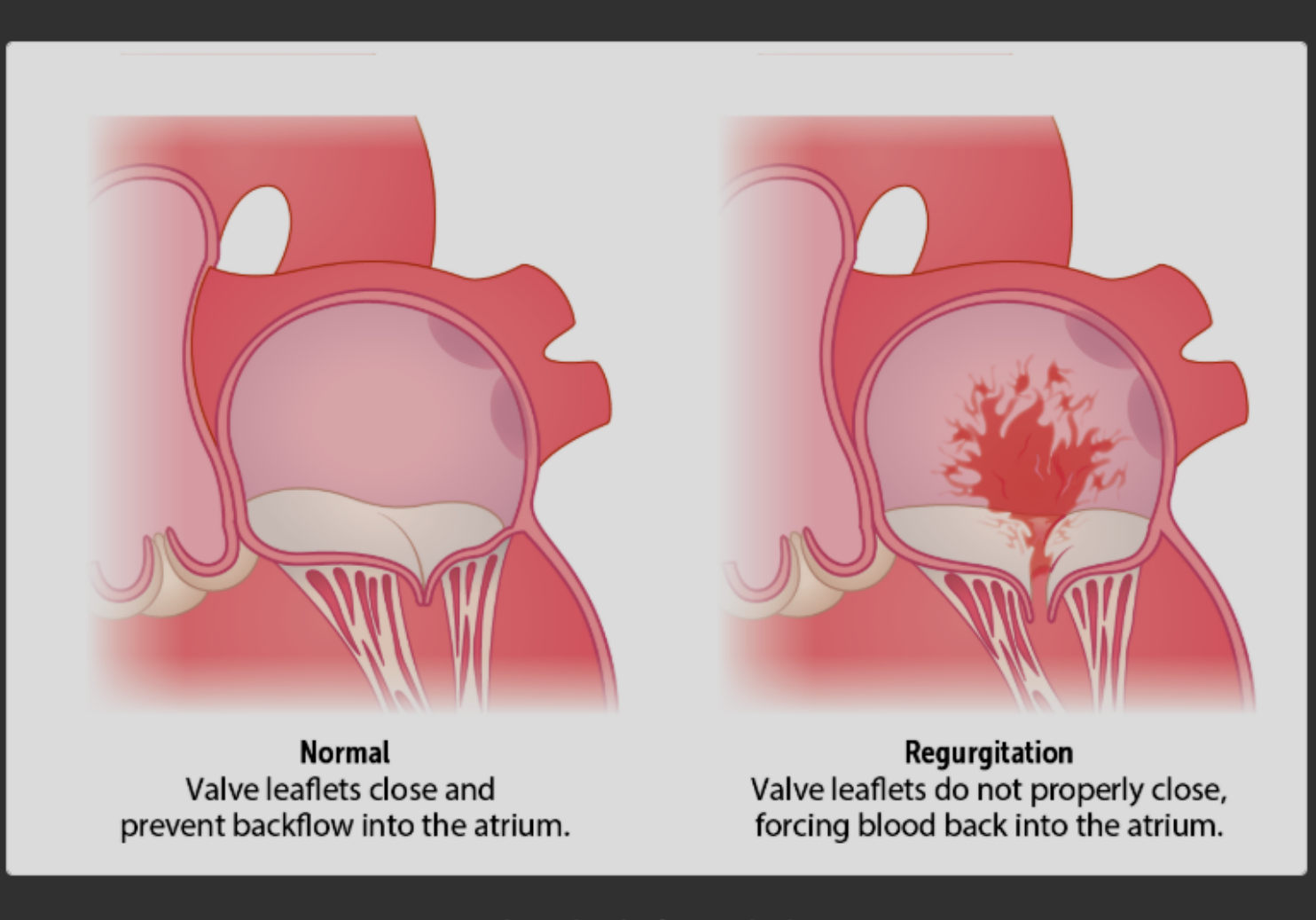

Normal
Valve leaflets close and prevent backflow into the atrium.
Regurgitation
Valve leaflets do not properly close, forcing blood back into the atrium.

Figure 2. Mitral regurgitation

MR is prevalent in 0.2% to 2% of people in the United States, which puts it as the second most common valvular heart disease following aortic stenosis. It is seen in as many as 10 dial Infarction (MI).

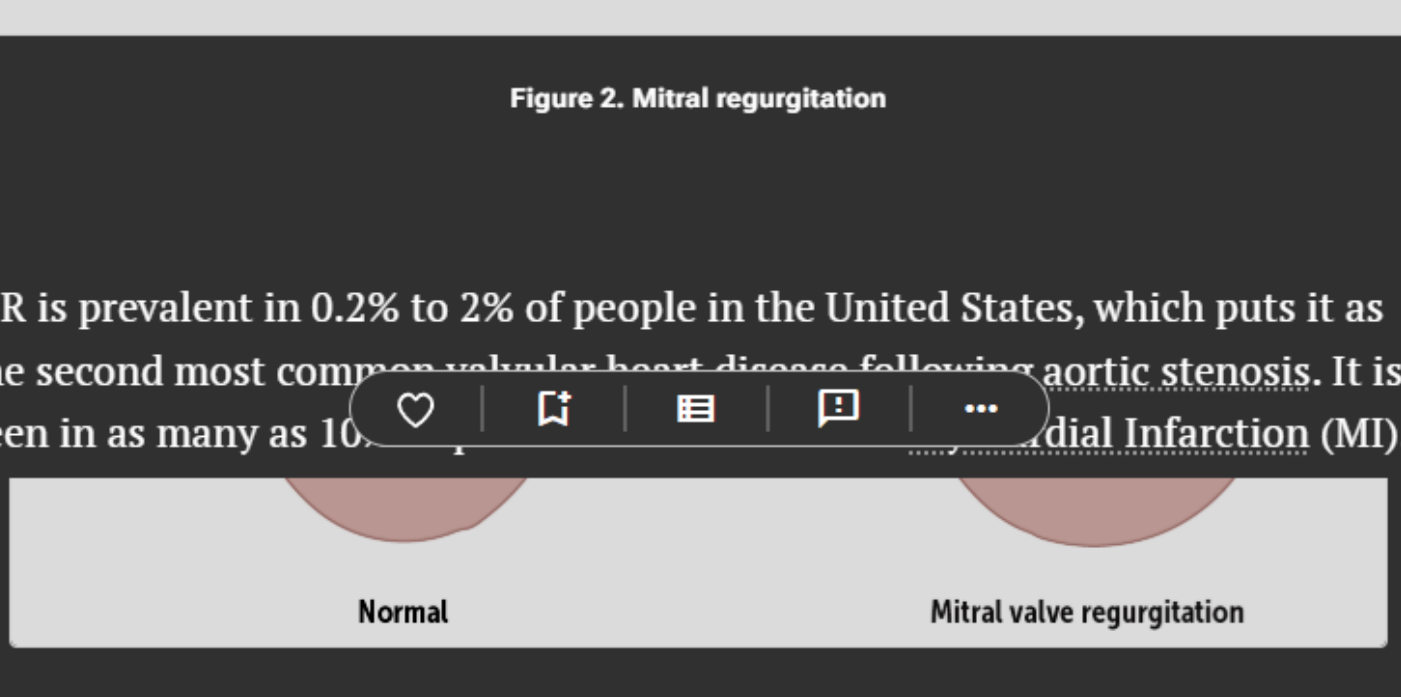


Figure 4. Tethering of mitral valve apparatus

How does myocardial ischemia cause acute valvular insufficiency?

Causes of Chronic Mitral Regurgitation

Chronic MR can be arise due to:

- **Myxomatous degeneration causing Mitral Valve Prolapse (MVP).** Disorganized connective tissue in the the middle (spongiosa) layer of the valve leaflets and in the chordae tendineae (Figure 5). Note that unlike the surrounding laye VP, the most common cause of Mitral Regurgitation in the US, is discussed further

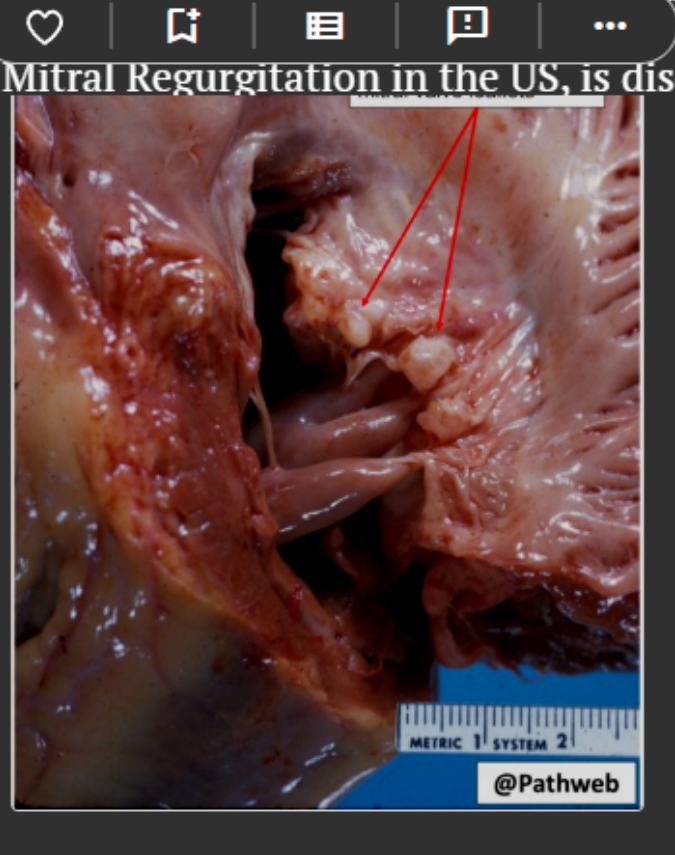
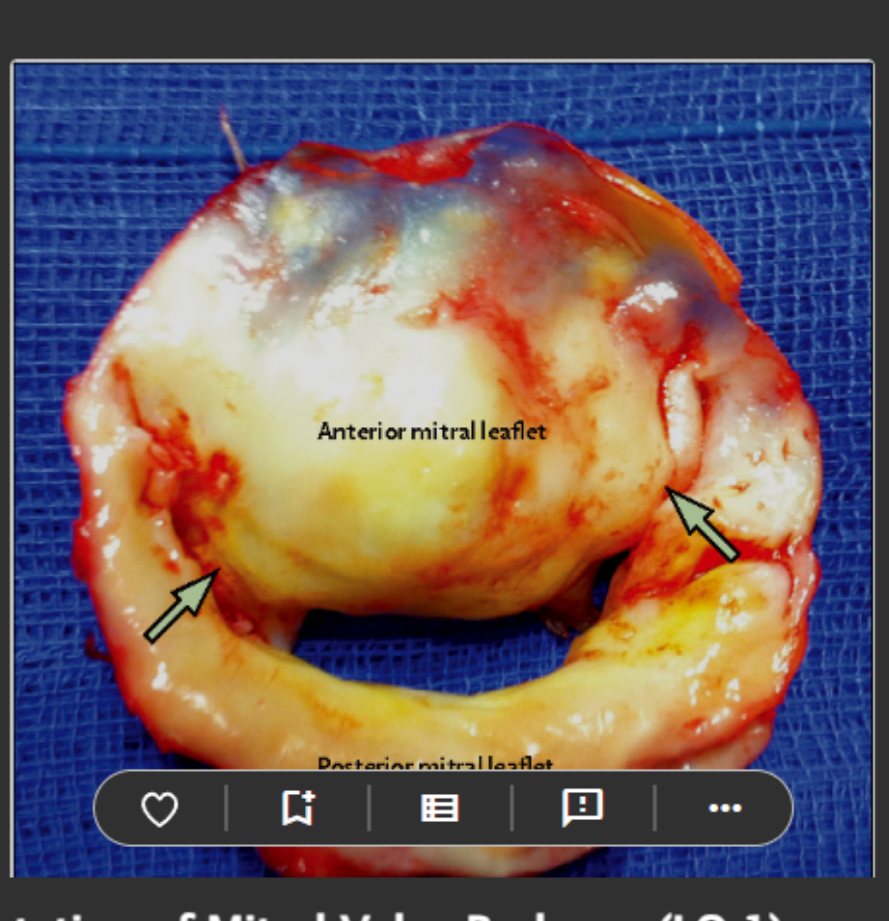


Figure 6. Infective Endocarditis of Mitral Valve



Clinical Presentation of Mitral Valve Prolapse (LO 1)

Most patients with MVP will have no symptoms unless they develop symptomatic MR. When patients with isolated MVP do have symptoms, they are nonspecific: shortness of breath, palpitations, exercise intolerance, dizziness, and anxiety. These symptoms were generally thought to be coincidental, yet a subset of patients with MVP may also have premature ventricular contractions and other dysrhythmias.

Etiology and Pathophysiology of Mitral Valve Prolapse (LO 2)

Mitral valve prolapse (MVP) is a condition in which the heart's mitral valve becomes floppy due to myxomatous degeneration, allowing it to bulge into the Left Atrium (LA) during ventricular systole. MVP is a common heart valve condition with an estimated prevalence of 2% to 3% of the population. It is more common in females than males, and more common in patients with lower body mass index.

The cause of myxomatous degeneration is unknown, but may be due to familial and sporadic gene mutations. This excess connective tissue causes the leaflets to thicken, the chordae tendineae to weaken, and the annulus to widen. This may initially cause Mitral Valve Prolapse (MVP). MVP progresses to MR in the *minority* of patients, yet **MVP is the most common cause of MR.**

Diagnosis of Mitral Valve Prolapse (LO 3)

Mitral Valve Prolapse is typically suspected because of a physical examination findings, a **mid-to-late systolic click** heard on auscultation. This click is believed to be caused dae tendineae when a

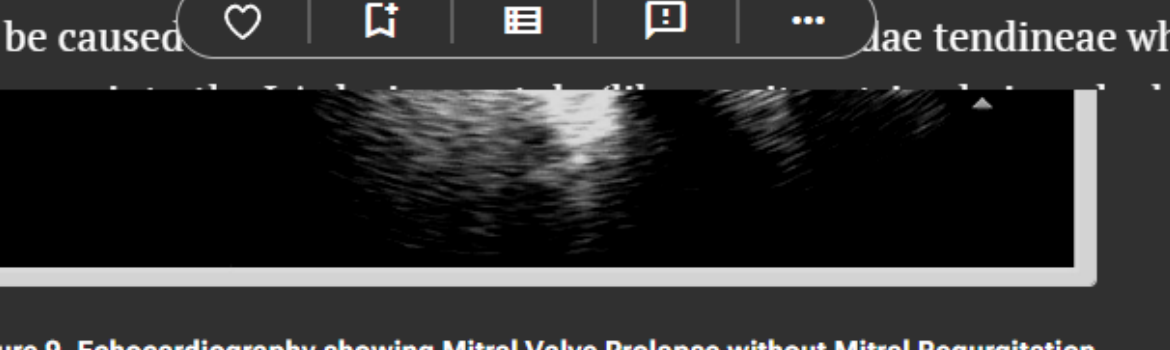


Figure 9. Echocardiography showing Mitral Valve Prolapse without Mitral Regurgitation

MVP progresses to MR in the minority of patients, estimated at 5-10% developing severe MR. When MR develops from MVP, it is heard as a **mid systolic click with late systolic murmur**.

Management of Mitral Valve Prolapse (LO 4)

MVP generally does not need treatment unless complications like mitral

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to RP, how do you proceed with his workup, and explain the expected findings?

You obtain an echocardiogram showing mitral valve prolapse with mitral regurgitation and mild left atrial enlargement. Degeneration of the mitral valve has progressed from prolapse to mitral regurgitation, and the blood leaks backward through the valve into the upper chamber and causes fluid buildup in the lungs. Also, because some of the blood leaks backward, less blood is supplied to the rest of the body. RP is prescribed an ACE inhibitor to decrease afterload. Regarding further treatment, you explain that the timing of mitral valve repair or replacement requires joint decision-making between clinician and patient and recommend a follow-up session for RP in 2 weeks.

Summary

- Mitral regurgitation (MR) is the abnormal leakage of blood from the left ventricle back into the left atrium during systole.
- Acute MR can be caused by myocardial infarction (papillary muscle rupture) or ischemia (tethering), infective endocarditis, or trauma.
- Chronic MR can be caused by valve abnormalities like progression of Mitral Valve Prolapse (MVP) with myxomatous degeneration of the valve mitral valve, chronic infective endocarditis, rheumatic heart disease, and congenital anomalies
- Chronic MR can a phy of the left

with which of the following?

A. ○ cardiac auscultation revealing a mid systolic click and early systolic murmur

B. ● slowly progressing symptoms of heart failure

C. ○ sudden, severe heart failure with no prior symptoms

D. ○ symptoms of heart failure as soon as MVP develops

Hide Explanation

The correct answer is (B). MVP, when it progresses, results in chronic MR, not acute (C). Most cases of MVP remain asymptomatic (D). The murmur of MR from MVP would occur after the prolapse/midsystolic click, as a late systolic murmur (A).

How would you rate this Brick?

References

- Douedi S, Douedi H. Mitral regurgitation. In: *StatPearls*.

What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Tricuspid Regurgitation and Tricuspid Stenosis

Last updated December 26, 2025 17 min read ScholarRx

Adapted by: Diptiman Bose, Patrick Brooks

Editors: Jordan Finn, Russell Johnson

Adapted from: Tricuspid Regurgitation and Tricuspid Stenosis

Listen to this Brick

0:00 / 27:01 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the etiologies and pathophysiology of tricuspid regurgitation.
2. Describe the clinical presentation, diagnosis, and management of tricuspid regurgitation.
3. Describe the uspid stenosis.

What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Tricuspid Regurgitation and Tricuspid Stenosis

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the etiologies and pathophysiology of tricuspid regurgitation.
2. Describe the clinical presentation, diagnosis, and management of tricuspid regurgitation.
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